Tensors Review:

$$t_\alpha{}^\beta{}_{\gamma\delta\varepsilon\zeta} = \eta_{\alpha\theta}\,\eta_{\gamma\tau}\,t^{\theta\beta\tau}{}_{\delta\varepsilon\zeta}$$

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$$

$$u^\mu = \begin{pmatrix} \gamma \\ \gamma\vec{\omega} \end{pmatrix} \qquad \eta_{\mu\nu}\,u^\nu = u_\mu = \begin{pmatrix} -\gamma \\ \gamma\vec{\omega} \end{pmatrix}$$

proper velocity

$$u^\mu u_\mu = -\gamma^2 + \gamma^2\vec{\omega}^2 = -\gamma^2(1-\vec{\omega}^2) = -1$$

$$p^\mu = m u^\mu$$

$$p^\mu p_\mu = -m^2$$

$$j^\mu = \begin{pmatrix} \rho \\ \rho\vec{\omega} \end{pmatrix} = \begin{pmatrix} \rho_0\gamma \\ \rho_0\gamma\vec{\omega} \end{pmatrix}$$

$$\partial_\mu j^\mu = 0 = \frac{\partial\rho}{\partial t} + \vec{\nabla}\cdot\vec{j}$$

$$\frac{\partial}{\partial x^\mu} = \begin{pmatrix} \frac{\partial}{\partial t} \\ \vec{\nabla} \end{pmatrix}$$

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{pmatrix}$$

$$\overset{\shortparallel}{-F^{\nu\mu}}$$

$$F^{0i} = E^i$$

$$F^{ij} = \epsilon^{ij}{}_k B^k$$

→ the electromagnetic field strength tensor

$$\Lambda = \begin{pmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

this is so nice!

$$F^{01} = \Lambda^0{}_\mu \Lambda^1{}_\nu F^{\mu\nu}$$

$$\overset{\shortparallel}{E_x'} = \cancel{(\Lambda^0{}_0 \Lambda^1{}_0 F^{00})}^{\,0} + \cancel{(\Lambda^0{}_0 \Lambda^1{}_1 F^{11})}^{\,0} + (\Lambda^0{}_1 \Lambda^1{}_0 F^{10}) + (\Lambda^0{}_0 \Lambda^1{}_1 F^{01})$$

$$= \gamma^2 E_x + (-\gamma v)^2 (-E_x)$$

$$= \gamma^2 E_x [1 - v^2]$$

$$= \frac{1-v^2}{1-v^2} E_x$$

$$= E_x.$$

$$F'^{02} = \Lambda^0{}_\mu \Lambda^2{}_\nu F^{\mu\nu}$$

$$\overset{\shortparallel}{E_y'} = (\Lambda^0{}_0 \Lambda^2{}_2) F^{02} + (\Lambda^0{}_1 \Lambda^2{}_2) F^{12}$$

$$= \gamma E_y + (-\gamma v) B_z$$

$$= \gamma (E_y - v B_z)$$

$$\tilde{F}^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$$

dual tensor

swaps $\vec{B}$ & $\vec{E}$

$\vec{B} \leftrightarrow -\vec{E}$

why?

$$\tilde{F} = \left(\begin{array}{c|c} & -B \\ \hline -B & \begin{matrix} 0 & -E_z & +E_y \\ & 0 & \\ & & 0 \end{matrix} \end{array} \right) \ldots$$

$$\left.\begin{aligned} \partial_\nu F^{\mu\nu} &= j^\mu \\ \partial_\nu \tilde{F}^{\mu\nu} &= 0 \end{aligned}\right\} \longrightarrow \text{Maxwell's equations.}$$

$$\vec{\nabla} \cdot \vec{E} = \rho$$

$$\nabla \times \vec{B} = \frac{\partial \vec{E}}{\partial t} + \vec{J}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}.$$

Maxwell's Equations are Lorentz Invariant.

$$\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$$

$$F^{\mu\nu} = \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & E_x & E_y & E_z \\ 1 & -E_x & 0 & B_z & \boxed{-B_y} \\ 2 & -E_y & -B_z & 0 & B_x \\ 3 & -E_z & \boxed{B_y} & -B_x & 0 \end{array}$$

$$= -F^{\nu\mu}$$

$$\tilde{F}^{00} = \tilde{F}^{11} = \tilde{F}^{22} = \tilde{F}^{33} = 0$$

$$\tilde{F}^{01} = \frac{1}{2}\left(\epsilon^{01\rho\sigma} F_{\rho\sigma}\right) = \frac{1}{2}\left(\epsilon^{0123} F_{23} + \epsilon^{0132} F_{32}\right)$$

$$= \frac{1}{2}\left(1 \cdot F_{23} - F_{32}\right) = \frac{1}{2}\left(B_x + B_x\right) = \frac{2B_x}{2} = B_x$$

$$\tilde{F}^{02} = \frac{1}{2}\left(\epsilon^{2\rho\sigma} F_{\rho\sigma}\right) = \frac{1}{2}\left(\epsilon^{0213} F_{13} + \epsilon^{0231} F_{31}\right)$$

$$= \frac{1}{2}\left(-F_{13} + F_{31}\right) = \frac{1}{2}\left(+B_y + B_y\right) = \frac{2B_y}{2} = B_y$$

$$\tilde{F}^{12} = \frac{1}{2}\left(\epsilon^{12\rho\sigma} F_{\rho\sigma}\right) = \frac{1}{2}\left(\epsilon^{1203} F_{03} + \epsilon^{1230} F_{30}\right)$$

$$= \frac{1}{2}\left[F_{03} + F_{30}\right] = \frac{1}{2}\left[E_z + E_z\right] = E_z.$$

$$\begin{array}{ccccccc} 0123 & \to & 1023 & \to & 1203 & \to & 1230 \\ +1 & & -1 & & +1 & & -1 \end{array}$$

turns out you can't calculate it

$$\epsilon_{\mu\nu\rho\sigma} \quad \epsilon_{0123} = +1 \quad \begin{cases} +1 & \text{even swaps} \\ -1 & \text{odd swaps} \\ 0 & \text{otherwise} \end{cases}$$

$$\epsilon^{0123} = g_0^{\alpha} g_1^{\beta} g_2^{\delta} g_3^{\gamma} \epsilon_{\alpha\beta\delta\gamma}$$

$$= \underset{\text{now}}{g_0^{0}} \overset{\text{does colm.}}{\square} \epsilon_{0\beta\delta\gamma} + \cancel{g_0^{1} \overset{0}{\square}} \epsilon_{1\beta\delta\gamma} + \cancel{g_0^{2} \overset{0}{\square}} \cdots$$

$$= (-1)\, g_1^{\beta} g_2^{\delta} g_3^{\gamma} \epsilon_{0\beta\delta\gamma}$$

$$= (-1)\, g_1^{1} g_2^{\delta} g_3^{\gamma} \epsilon_{01\delta\gamma}$$

$$= -\epsilon_{0123}. \quad \star$$

← the most important thing is to remember that $\epsilon^{\alpha\beta\rho\sigma}$ is a tensor!!

$$\tilde{F}^{00} = \tilde{F}^{11} = \tilde{F}^{22} = \tilde{F}^{33} = 0$$

$$\tilde{F}^{01} = \frac{1}{2}\left[\epsilon^{0123} F_{0123} + \epsilon^{0132} F_{0132}\right]$$

$$= \frac{1}{2}\left[(-1)(B_x) + (1)\cdot(-B_x)\right] = -B_x \quad \checkmark\checkmark$$

Hence the E & $\vec{B}$ exchange places with flipped signs.

$$\tilde{F}^{\mu\nu} = \left[\begin{array}{c|ccc} 0 & -B_x & -B_y & -B_z \\ \hline B_x & 0 & -E_z & +E_y \\ B_y & +E_z & 0 & -E_x \\ B_z & -E_y & +E_x & 0 \end{array}\right]$$

this should be lower index, I believe?

$$F^{\mu\nu} = -F^{\nu\mu} \qquad \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & E_x & E_y & E_z \\ 1 & -E_x & 0 & B_z & \boxed{-B_y} \\ 2 & -E_y & -B_z & 0 & B_x \\ 3 & -E_z & \boxed{B_y} & -B_x & 0 \end{array}$$

Suppose we live in D-spatial dimensions.

$$F^{\mu\nu} = \left(\begin{array}{c|cccc} & \vec{E} & & & \\ \hline -\vec{E} & 0 & B^{12} & B^{13} & B^{14} \cdots B^{1d} \\ & & \ddots & & \\ & & & \ddots & \end{array}\right) \Bigg\} \frac{d(d-1)}{2} \text{ components of } B.$$

$$F^{\mu\nu},\ \tilde{F}^{\mu\nu},\ \eta_{\mu\nu},\ \epsilon_{\mu\nu\rho\sigma}$$

$$\tilde{F}^{\mu\nu}\tilde{F}_{\mu\nu} = \begin{cases} F^{\mu\nu}F_{\mu\nu} \propto E^2 - B^2 . \\ \tilde{F}^{\mu\nu}F_{\mu\nu} \propto \vec{E}\cdot\vec{B} . \end{cases}$$

$\rightarrow$ these are invariant quantities

$$\frac{\partial \rho}{\partial t} = -\vec{\nabla}\cdot\vec{J}$$

charge conservation makes sense locally.

$$\tilde{F}^{\mu\nu} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \left[\begin{array}{c|ccc} 0 & -B_x & -B_y & -B_z \\ \hline B_x & 0 & -E_z & +E_y \\ B_y & +E_z & 0 & -E_x \\ B_z & -E_y & +E_x & 0 \end{array}\right] \end{array}$$

$$F^{\mu\nu} = -F^{\nu\mu} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} & 1 & 2 & 3 \end{array} \\ \left(\begin{array}{c|ccc} 0 & E_x & E_y & E_z \\ \hline -E_x & 0 & B_z & \boxed{-B_y} \\ -E_y & -B_z & 0 & B_x \\ -E_z & \boxed{B_y} & -B_x & 0 \end{array}\right) \end{array}$$

Verification:

$$\tilde{F}^{0\nu} F_{0\nu} = \tilde{F}^{00} F_{00} + \tilde{F}^{01} F_{01} + \tilde{F}^{02} F_{02} + \tilde{F}^{03} F_{03}$$

$$= 0 + (-B_x)(E_x) + (-B_y)(E_y) + (-B_z)(E_z)$$

$$= -B_x E_x - B_y E_y - B_z E_z$$

$$\tilde{F}^{1\nu} F_{1\nu} = \tilde{F}^{10} F_{10} + \tilde{F}^{11} F_{11} + \tilde{F}^{12} F_{12} + \tilde{F}^{13} F_{13}$$

$$= (B_x)(-E_x) + 0 + (-E_z)(B_z) + (E_y)(-B_y)$$

$$= -B_x E_x - B_z E_z - B_y E_y$$

$$\tilde{F}^{2\nu} F_{2\nu} = \tilde{F}^{20} F_{20} + \tilde{F}^{21} F_{21} + \cancel{\tilde{F}^{22} F_{22}} + \tilde{F}^{23} F_{23}$$

$$= B_y(-E_y) + E_z(-B_z) + (-E_x) B_x$$

$$= -B_y E_y - E_z B_z - E_x B_x$$

$$\tilde{F}^{3\nu} F_{3\nu} = \tilde{F}^{30} F_{30} + \tilde{F}^{31} F_{31} + \tilde{F}^{32} F_{32} + \cancel{\tilde{F}^{33} F_{33}}$$

$$= (B_z)(-E_z) + (-E_y)(B_y) + (E_x)(-B_x)$$

$$= -E_x B_x - E_y B_y - E_z B_z$$

$$\underline{\tilde{F}^{\mu\nu} F_{\mu\nu} = -4(\vec{B} \cdot \vec{E})}$$ cool!

which one should you place on the left? α or μ?

$$F^{\mu\nu} = g^{\alpha}_{\mu}\, g^{\beta}_{\nu}\, F_{\alpha\beta}$$

$$F^{00} = g_0^{\ \alpha}\, g_0^{\ \beta}\, F_{\alpha\beta}$$

all the other term sum to 0.

$$= g_0^{\ 0}\, g_0^{\ 0}\, F_{00} + \cancel{g_0^{\ \alpha}\, g_0^{\ \beta}\, F_{\alpha\beta}}$$

$$F^{01} = g_0^{\ \alpha}\, g_1^{\ \beta}\, F_{\alpha\beta} = g_0^{\ 0}\, g_1^{\ 1}\, F_{01} = (-1)(1)(E_x) = -E_x$$

so I guess any term involving 0 will swap sign, but the other terms won't change.

$$F^{\mu\nu} = \left[\begin{array}{c|ccc} 0 & -E_x & -E_y & -E_z \\ \hline E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & +B_y & -B_x & 0 \end{array}\right]$$

won't the signs are so confusing.

$$F^{00}F_{00} = 0, \quad F^{01}F_{01} = -E_x^2, \quad F^{02}F_{02} = -E_y^2, \quad F^{03}F_{03} = -E_z^2$$

$$F^{10}F_{10} = -E_x^2, \quad F^{20}F_{20} = -E_y, \quad F^{30}F_{30} = -E_z^2$$

$$F^{12}F_{12} = B_z^2 \ \ldots\ldots$$

Then, sum to $2\left[(B_x^2 + B_y^2 + B_z^2) - (E_x^2 + E_y^2 + E_z^2)\right]$

$$= 2\left[B^2 - E^2\right]$$

Supplementary Notes:

Maxwell's Equations:

① Gauss's Law for Electricity:

$$\nabla \cdot \vec{E} = \frac{\partial E}{\partial x} + \frac{\partial E}{\partial y} + \frac{\partial E}{\partial z} = \frac{\rho}{\varepsilon_0}$$

Electric charges are sources/sinks of electric field.

② Gauss's law for Magnetism

$$\nabla \cdot \vec{B} = \frac{\partial B}{\partial x} + \frac{\partial B}{\partial y} + \frac{\partial B}{\partial z} = 0.$$

There are no magnetic monopoles.
At every point in space, as much magnetic field flows in as flows out.

③ Faraday's law of Induction

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = \begin{pmatrix} \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \\ \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \\ \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \end{pmatrix}$$

A changing magnetic field produces an electric field.

④ Ampère-Maxwell law

$$\nabla \times \vec{B} = \begin{pmatrix} \frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \\ \frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \\ \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \end{pmatrix} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Magnetic fields are created by moving charges & changing electric fields.

Relativistic / Tensor Form.

$$F^{\mu\nu} = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & +B_y & -B_x & 0 \end{bmatrix}$$

① Inhomogenous equations: (sources):

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$$

$$\partial_\mu F^{\mu 0} = \partial_0 F^{00} + \partial_1 F^{10} + \partial_2 F^{20} + \partial_3 F^{30}$$

$$= 0 + \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = \mu_0 \rho$$

$$\nabla \cdot E = \mu_0 \rho$$

since $\mu_0 \varepsilon_0 c^2 = 1$

$$= \frac{\rho}{\varepsilon_0} \rightarrow \text{Gauss's Law.}$$

$$\partial_\mu F^{\mu 1} = \partial_0 F^{01} + \partial_1 F^{11} + \partial_2 F^{21} + \partial_3 F^{31}$$

$$= -\frac{\partial E_x}{\partial t} + 0 - \frac{\partial B_z}{\partial y} + \frac{\partial B_y}{\partial z} = \mu_0 J_x$$

$$\ldots$$

$\partial_\mu F^{\mu 2}$

$\partial_\mu F^{\mu 3}$

$$\boxed{\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}}$$

Ampère - Maxwell Law.

② Homogenous Eqn.

$$\partial_\nu \tilde{F}^{\mu\nu} = 0$$

$$\tilde{F}^{\mu\nu} = \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & -B_x & -B_y & -B_z \\ 1 & B_x & 0 & -E_z & +E_y \\ 2 & B_y & +E_z & 0 & -E_x \\ 3 & B_z & -E_y & +E_x & 0 \end{array}$$

$$\partial_\nu \tilde{F}^{0\nu} = \partial_0 \tilde{F}^{00} + \partial_1 \tilde{F}^{01} + \partial_2 \tilde{F}^{02} + \partial_3 \tilde{F}^{03}$$

$$= 0 + \frac{\partial(-B_x)}{\partial x} + \frac{\partial(-B_y)}{\partial y} + \frac{\partial(-B_z)}{\partial z}$$

$$= -\nabla \cdot \vec{B} = 0 \Rightarrow \boxed{\nabla \cdot \vec{B} = 0} \rightarrow \text{Gauss's law.}$$

$$\partial_\nu \tilde{F}^{1\nu} = \partial_0 \tilde{F}^{10} + \cancel{\partial_1 \tilde{F}^{11}} + \partial_2 \tilde{F}^{12} + \partial_3 \tilde{F}^{13}$$

$$= \frac{\partial B_x}{\partial t} + \frac{\partial(-E_z)}{\partial y} + \frac{\partial(E_y)}{\partial z} = 0$$

$$\partial_\nu \tilde{F}^{2\nu}$$

$$\partial_\nu \tilde{F}^{3\nu}$$

$$\boxed{-\frac{\partial B}{\partial t} = \nabla \times \vec{E}} \rightarrow \text{Faraday's law of induction.}$$

that's very cool!

$p^\mu = \begin{pmatrix} E \\ \vec{p} \end{pmatrix}$,

Energy-Momentum Tensor: $T^{\mu\nu}$ → gravitational source.

energy-momentum tensor.

[Symmetric: $T^{\mu\nu} = T^{\nu\mu}$].

$$T^{\mu\nu} = \begin{pmatrix} T^{00} & T^{01} & T^{02} & T^{03} \\ T^{10} & T^{11} & T^{12} & T^{13} \\ T^{20} & T^{21} & T^{22} & T^{22} \\ T^{30} & T^{31} & T^{32} & T^{33} \end{pmatrix}$$

energy density.

flow of energy through space.

flux of energy.

$\frac{dP}{dt}$

pressure

momentum density.

flux of momentum through space.

how momentum changes with position.

$$\partial_\nu T^{\mu} = 0$$

$$\partial_\nu T^{0\nu} = 0 \rightarrow \text{energy conservation.}$$

$$\partial_\nu T^{i\nu} = 0 \rightarrow \text{momentum conservation}$$

T^{ij} is the flux of p^i through a surface $\perp \hat{x}^j$.

$$T^{\mu\nu} = \begin{pmatrix} \rho & & & \\ & P & & \\ & & P & \\ & & & P \end{pmatrix}$$

no heat conduction.

no viscosity (no shear forces).

ideal fluid

Suppose the fluid moves with u^μ proper velocity

In rest frame, $\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

$$T^{00} = \alpha - \beta, \quad T^{11} = T^{22} = T^{33} = \beta.$$

$$\boxed{\begin{aligned} T^{\mu\nu} &= \alpha u^\mu u^\nu + \beta \eta^{\mu\nu} \\ &= (\rho + p) u^\mu u^\nu + p \eta^{\mu\nu} \end{aligned}}$$

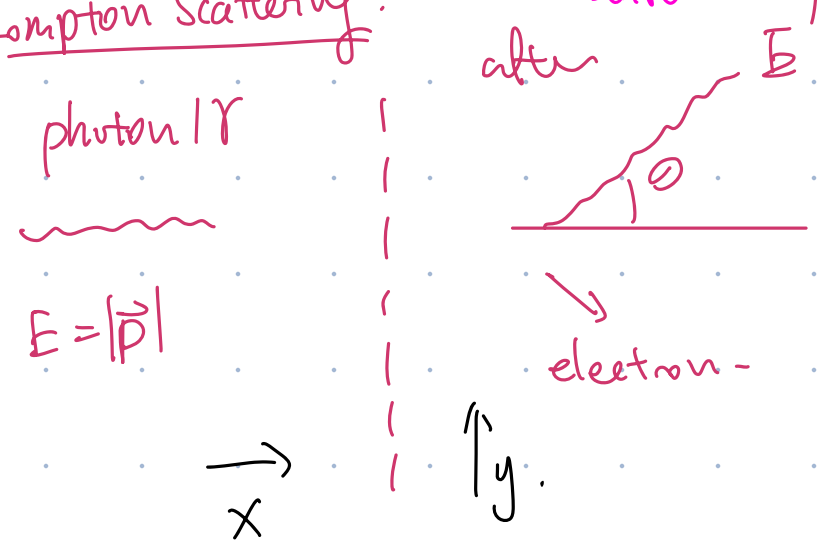
$$\begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$$

$$\sum_{\text{before}} p_i^\mu = \sum_{\text{after}} p_j^\mu \qquad \{i\} \neq \{j\}.$$

$$p_i^\mu p_{i,\mu} = -m_i^2 \quad , \quad E^2 - \vec{p}^2 = m^2$$

for $m = 0$, $E = |\vec{p}|$.

Compton Scattering: → photon bounces off electron & scatters @ an angle.



$$P^{\mu}_{\gamma,b} = \begin{pmatrix} E \\ E \\ 0 \\ 0 \end{pmatrix}, \quad P^{\mu}_{e,b} = \begin{pmatrix} m_e \\ 0 \\ 0 \\ 0 \end{pmatrix},$$

before

photon

derived based on momentum conservation.

$$P^{\mu}_{\gamma,a} = \begin{pmatrix} E' \\ E'\cos\theta \\ E'\sin\theta \\ 0 \end{pmatrix} \quad P^{\mu}_{e,a} = \begin{pmatrix} m_e + E - E' \\ E - E'\cos\theta \\ -E'\sin\theta \\ 0 \end{pmatrix}$$

$$-m_e^2 = P^{\mu}_{e,a}(P_{e,a})_{\mu}$$ → invariance of momentum

$$= -(m_e + E + E')^2 + (E - E'\cos\theta)^2 + (-E'\sin\theta)^2$$

$$-m_e^2 = -\left[(m_e^2 + E^2 + 2m_eE) + E'^2 + 2(m_e + E)E'\right] + \left[E^2 + E'^2\cos^2\theta - 2EE'\cos\theta\right] + E'^2\sin^2\theta$$

$$\cancel{-m_e^2} = \cancel{-m_e^2} \cancel{-E^2} - 2m_eE \cancel{-E'^2} - 2m_eE' \cancel{-2m_eE'} \cancel{+E^2} + \cancel{(E'^2\cos^2\theta + E'^2\sin^2\theta)} - 2EE'\cos\theta$$

$2EE'$

...

$$m_e(E + E') = EE'(1 - \cos\theta)$$

$$\frac{1}{E'} - \frac{1}{E} = \frac{1 - \cos\theta}{m_e}.$$